

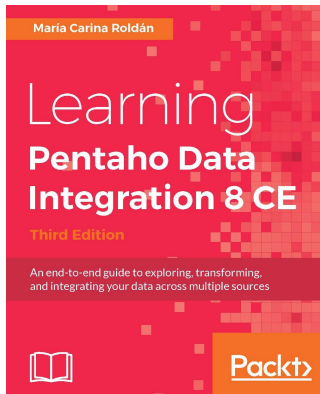
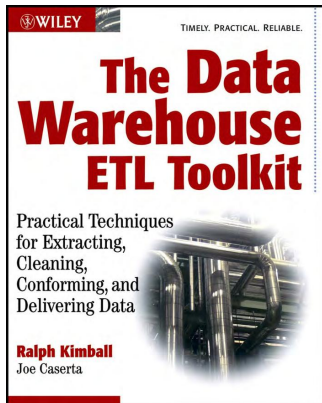
Database Administration

Lecture 05: ETL – Extraction, Transformation & Load

Kimball, Caserta & Roldán

30 de agosto de 2026

Database Administration: ETL – Extraction, Transformation & Load.



Content has been extracted from “*The Data Warehouse ETL Toolkit*” by Kimball & Caserta, 2004. Visit kimballgroup.com and “*Learning Pentaho Data Integration 8 CE*” by Roldán, 2018. Visit oreilly.com.

Plan

DW Intro

Main Requirements

Categories of Requirements

Architectural Decisions

Back Room & Front Room

Mission

What is PDI?

What can you do with PDI?

Meet Spoon

Your first Transformation

What is a Data Warehouse?

- ▶ **Purpose:** Publish organizational data assets to support decision-making.
- ▶ **Core traits:** Subject/process-oriented, integrated, time-variant, non-volatile.
- ▶ **Users:** Analysts, managers, apps (dashboards, reports, OLAP, data science).
- ▶ **Outcomes:** Trusted metrics, faster insight, single version of the truth.

Main Components (Kitchen & Dining Metaphor)

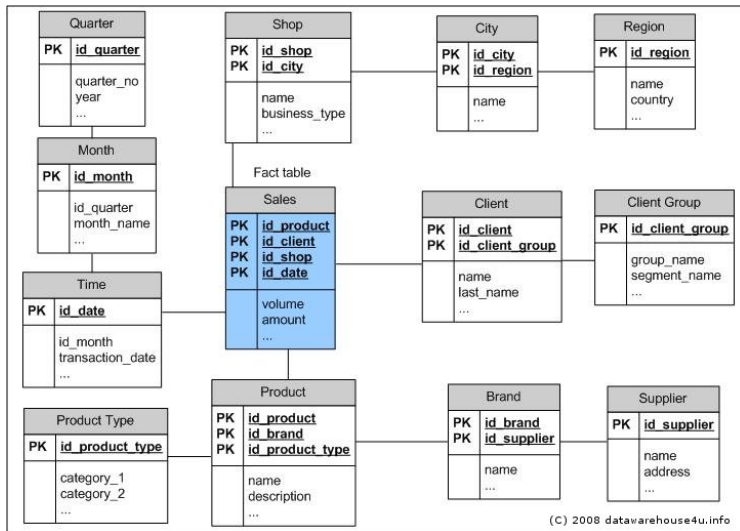
Back Room (Kitchen)

- ▶ Staging & processing area (no end-user access)
- ▶ Raw sources → standardized, high-quality data
- ▶ Governance: lineage, quality checks, security, restart/recovery

Front Room (Dining room)

- ▶ Dimensional models / data marts (atomic + aggregates)
- ▶ Access by BI tools: SQL, reports, dashboards, OLAP, ML
- ▶ Performance tuning, semantic consistency (*conformed* dims/facts)

DW Schema



ETL Responsibilities

- ▶ Add value: quality, standard units, surrogate keys, SCDs
- ▶ Preserve lineage & auditability; archive staging
- ▶ Automate: scheduling, exceptions, recovery/restart
- ▶ Meet **latency** needs: batch or streaming (real-time)

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Why Requirements Matter

- ▶ ETL design begins with surrounding the requirements.
- ▶ Requirements are non-negotiable constraints to adapt to.
- ▶ Early architectural decisions drive:
 - ▶ Hardware & software
 - ▶ Coding practices
 - ▶ Personnel & operations
- ▶ Clear mission: define back room, staging, operational data stores, presentation area.

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Business Needs

- ▶ End users' information requirements.
- ▶ Business needs drive the choice of data sources.
- ▶ Interviews and investigations often uncover:
 - ▶ Hidden complexities or limitations
 - ▶ Additional capabilities of data sources
- ▶ Continuous dialogue between ETL team, architects, and end users.

Compliance & Security

- ▶ Sarbanes-Oxley and other regulations demand:
 - ▶ Proof of accuracy, completeness, and lineage.
 - ▶ Archived copies and documented algorithms.
- ▶ Security:
 - ▶ Role-based access control (via directory server).
 - ▶ Separate ETL subnets, controlled backups, logs.

Other Requirements

- ▶ **Data Profiling:** assess quality, completeness, and usability.
- ▶ **Integration:** conforming dimensions & facts.
- ▶ **Latency:** batch vs streaming delivery.
- ▶ **Archiving & Lineage:** keep staged data + metadata.
- ▶ **End User Interfaces:** responsibility of ETL to simplify delivery.
- ▶ **Skills & Legacy:** staff expertise and existing licenses impact design.

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ETL Tool vs. Hand Coding

ETL Tool Advantages:

- ▶ Faster development, metadata management.
- ▶ Built-in scheduling, connectors, lineage tracking.
- ▶ Good performance at scale.

Hand-Coded Advantages:

- ▶ Unlimited flexibility, OOP, unit testing.
- ▶ Full control over metadata.
- ▶ Avoid vendor lock-in.

Other Architectural Issues

- ▶ Proven technology vs. untested tools.
- ▶ Batch vs. Streaming data flows.
- ▶ Task dependency: Horizontal vs. Vertical (latency vs consistency).
- ▶ Scheduler automation and monitoring.
- ▶ Exception handling, quality management, recovery.
- ▶ Metadata repositories and process-flow tracking.

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The Back & Front Rooms

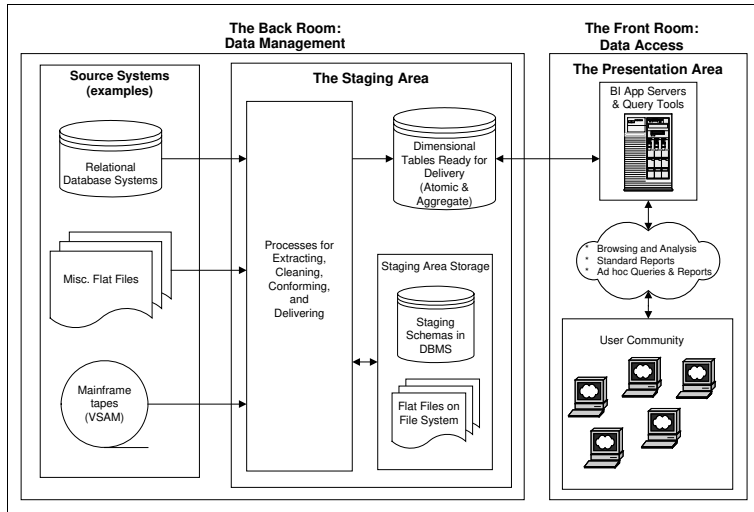


Figure 1.1 The back room and front room of a data warehouse.

The Back Room – Data Management

- ▶ Kitchen metaphor: preparation behind the scenes.
- ▶ Four staging steps:
 1. Extract
 2. Clean
 3. Conform
 4. Deliver
- ▶ Strictly off-limits to end users.

Four staging steps

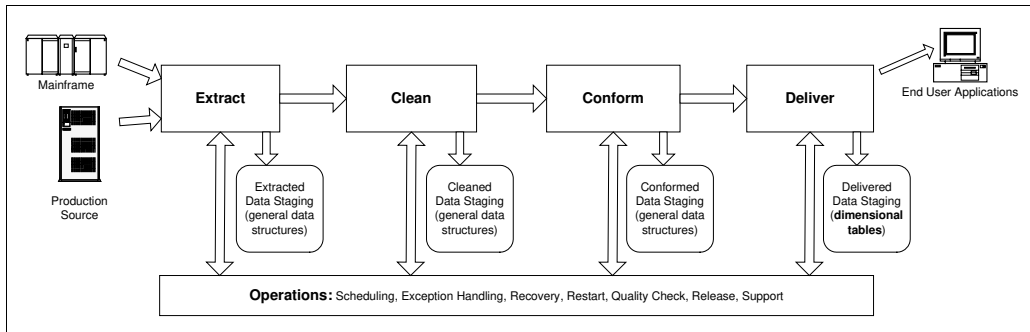


Figure 1.2 The Four Staging Steps of a Data Warehouse.

The Front Room – Data Access

- ▶ Presentation layer for queries, dashboards, OLAP cubes.
- ▶ Data marts = measurement-intensive subject areas.
- ▶ Must be atomic (detail) + aggregated (pyramidal).
- ▶ Can be centralized or decentralized, but always conformed.

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Mission of the Data Warehouse & ETL Team

Data Warehouse:

- ▶ Publish data assets to support decision-making.
- ▶ Deliver reliable, usable, and timely information.

ETL Team:

- ▶ Build the back room.
- ▶ Add value by cleaning and conforming data.
- ▶ Protect and document lineage.
- ▶ Deliver data for querying, reporting, dashboards.

Closing Thoughts

- ▶ Requirements are the foundation of ETL design.
- ▶ Early architecture decisions shape the entire system.
- ▶ Back room = preparation; Front room = access.
- ▶ The ETL team's mission is strategic to DW success.

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Pentaho Data Integration in a Nutshell

- ▶ **PDI (a.k.a. Kettle)** = engine + tools for **Extract, Transform, Load (ETL)**.
- ▶ Part of the **Pentaho BI Suite**: analysis (Mondrian), reporting, data mining (Weka), dashboards (CTools), *etc.*
- ▶ Tight platform services: *authz/authn, scheduling, web services, scalability, failover.*
- ▶ Community roots → adopted by Pentaho; rapid evolution with frequent releases.

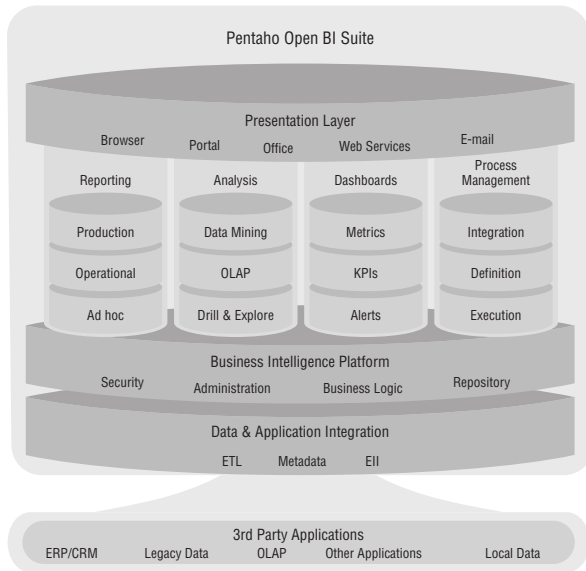
Where PDI fits in the BI stack

- ▶ **Data Integration (PDI)** feeds:
 - ▶ OLAP (Mondrian)
 - ▶ Reports (*Pentaho Reporting*)
 - ▶ Data mining (Weka, R/CPython steps)
 - ▶ Dashboards (CDE/CCC/CDA)
- ▶ Can run standalone or embedded in the platform.

Key benefits

- ▶ Open source ecosystem
- ▶ Broad connectivity
- ▶ Visual design (Spoon)
- ▶ Scales from laptop to cluster

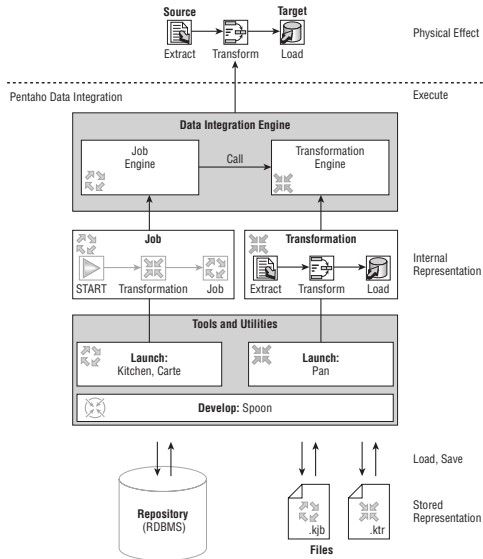
Pentaho Architecture



Characteristics

- ▶ Data integration: Combining data from different sources to provide a unified view.
- ▶ Pentaho Data Integration (PDI) offers tools for ETL (Extract, Transform, Load).
- ▶ Core PDI components: Transformations, Jobs, and the Data Integration Engine.

Data Integration Architecture



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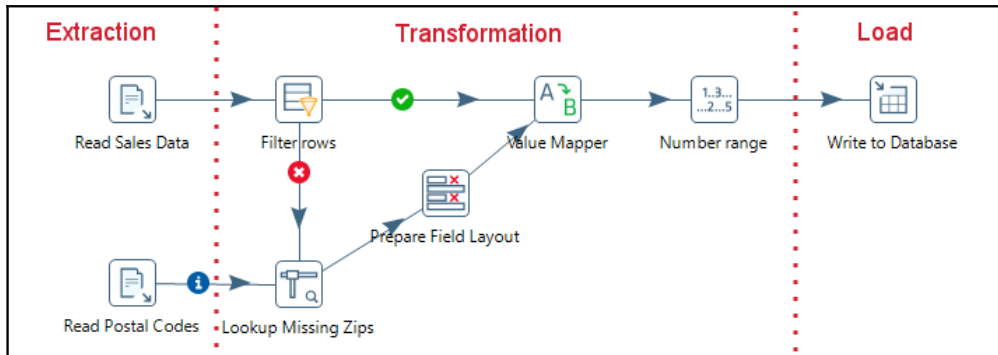
What can you do with PDI?

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Your first Transformation

Typical uses (beyond “just ETL”)

- **Load data warehouses/marts** (E → T → L pipelines, SCD handling).



Data Integration Activities

- ▶ Extraction: Retrieving data from various sources.
- ▶ Change Data Capture (CDC): Identifying changes in source data.
- ▶ Data Staging: Intermediate storage for transformation.
- ▶ Data Validation and Cleansing: Ensuring data quality.
- ▶ Key Management and Aggregation.
- ▶ Dimension and Fact Table Loading.

Typical uses (beyond “just ETL”)

- ▶ **Integrate systems** (ERP + CRM, mergers, multi-source unification).
- ▶ **Data cleansing** (validation, standardization, deduplication, defaults).
- ▶ **Migrations** (schemas/files $\leftrightarrow$ RDBMS/spreadsheets).
- ▶ **Exports & interoperability** (regulatory, inter-dept sharing).
- ▶ **Orchestration/automation** (emails, scheduled jobs, preprocess feeds for reports/dashboards).

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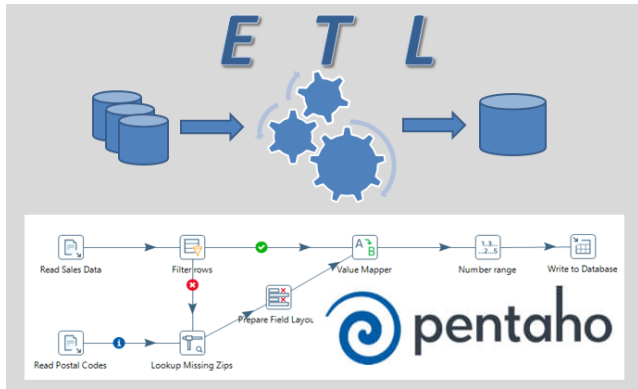
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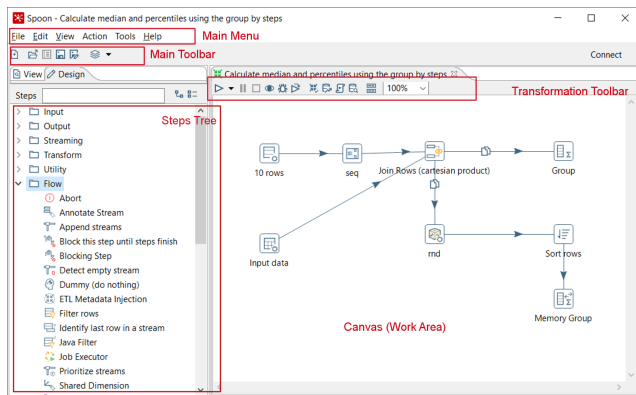
Pentaho Data Integration Components

- ▶ **Kettle**: Data integration engine.
- ▶ **Spoon**: Graphical IDE for designing transformations and jobs.
- ▶ **Kitchen**: Command-line tool for executing jobs.
- ▶ **Pan**: Command-line tool for running transformations.
- ▶ **Carte**: Remote execution engine.



Getting Started with Spoon

- ▶ Launch Spoon and create a new transformation.
- ▶ Add steps to extract, transform, and load data.
- ▶ Connect steps using “hops”.
- ▶ Preview and execute transformations.



Spoon interface at a glance

- ▶ **Main Menu/Toolbar**
- ▶ **Design** (Steps tree)
- ▶ **View** (structure/logs/metrics)
- ▶ **Canvas** (your pipeline graph)
- ▶ **Transformation Toolbar** (preview, run, debug)

Mental model

Transformation = *steps* + *hops*
(dataflow oriented).

Artifacts are metadata (XML)
interpreted by the Kettle engine.

Extending PDI with the Marketplace

- ▶ **Tools** → **Marketplace**: browse/install plugins by *Type* and *Maturity*.
- ▶ Two lanes: *Community* vs *Customer*; stages 1–4 (*lab* → *production-ready*).
- ▶ Some plugins are EE-only; descriptions indicate availability.

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Hello, World! (hands-on in 90 seconds)

1. **File** → **New** → **Transformation**.
2. Drag **Input** → **Data Grid** to canvas; define a **name** column and sample rows.
3. Drag **Scripting** → **User Defined Java Expression**; hop from Data Grid to UDJE.
4. In UDJE, create field `hello_message = ‘‘Hello, ’’ + name + ‘‘!’’`.
5. **Preview** (magnifier icon) on UDJE to sample output; then **Run**.

Save it

Edit → **Settings** → name, description, extended description → **File** → **Save**.

Summary

- ▶ PDI is a versatile, pluggable **data integration** engine with a visual designer.
- ▶ PDI provides tools for efficient ETL processes.
- ▶ Transformations process data at the record level.
- ▶ Jobs orchestrate multiple tasks.
- ▶ Spoon offers a user-friendly interface for development.
- ▶ You can **prototype quickly** (preview/run) and scale up as needed.
- ▶ Marketplace accelerates adoption via **plugins**—mind the maturity stage.

End of Lecture 5.



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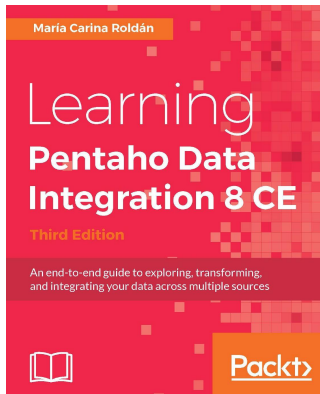
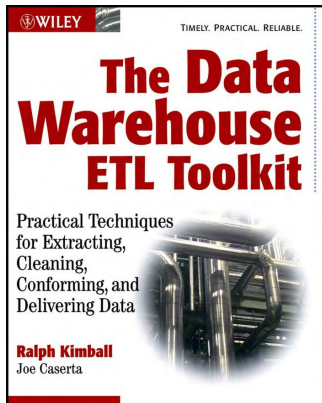
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- 2 A data warehouse employs a **two-component architecture**: a ‘**back room**’ dedicated to data management and preparation, and a ‘**front room**’ for user data access and analysis.
- 1 The **data warehouse’s central mission** is to **publish organizational data assets for effective decision-making**, with the **ETL team’s core task** being to build the ‘back room’ by cleaning, conforming, documenting lineage, and delivering data dimensionally.

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